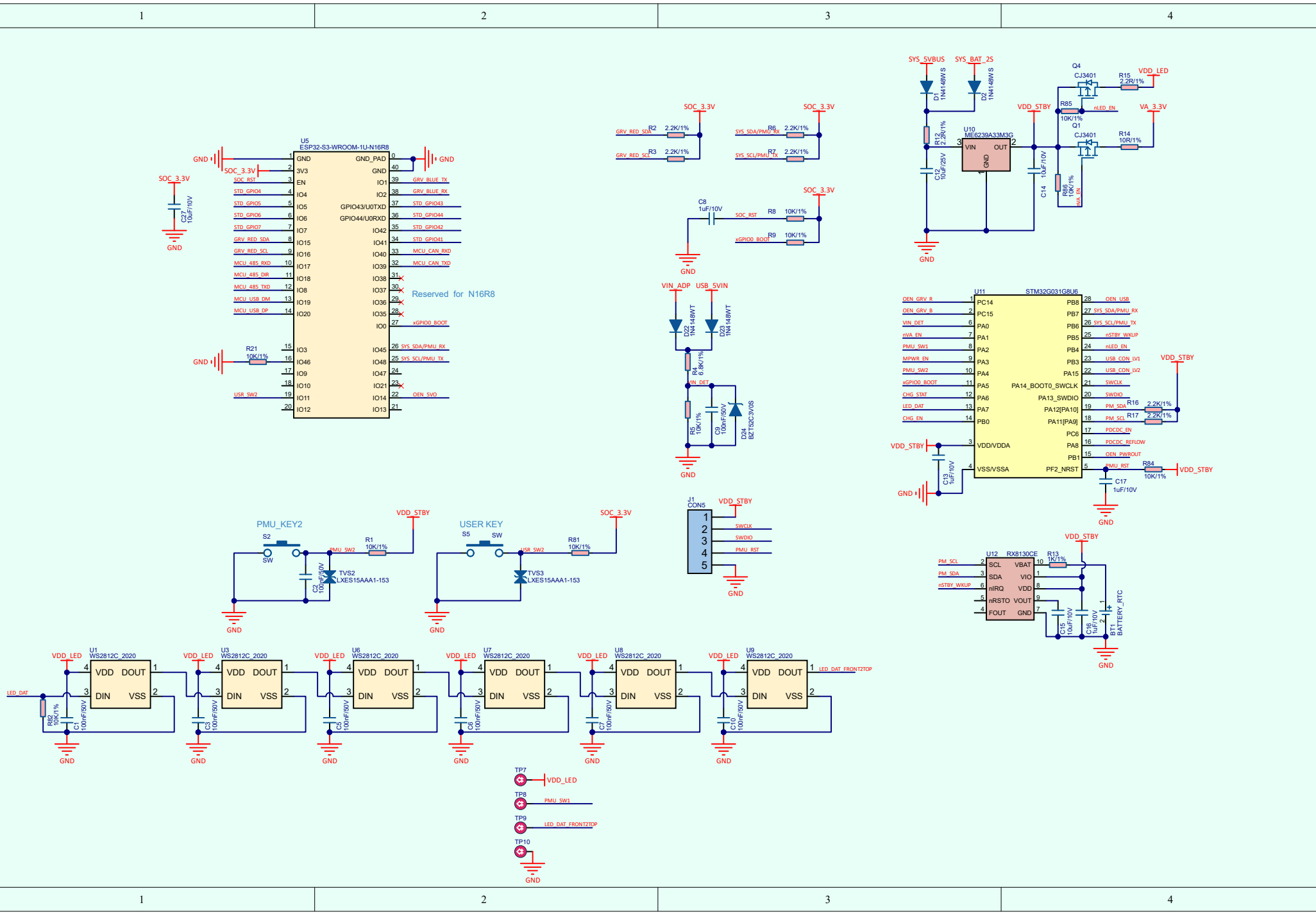
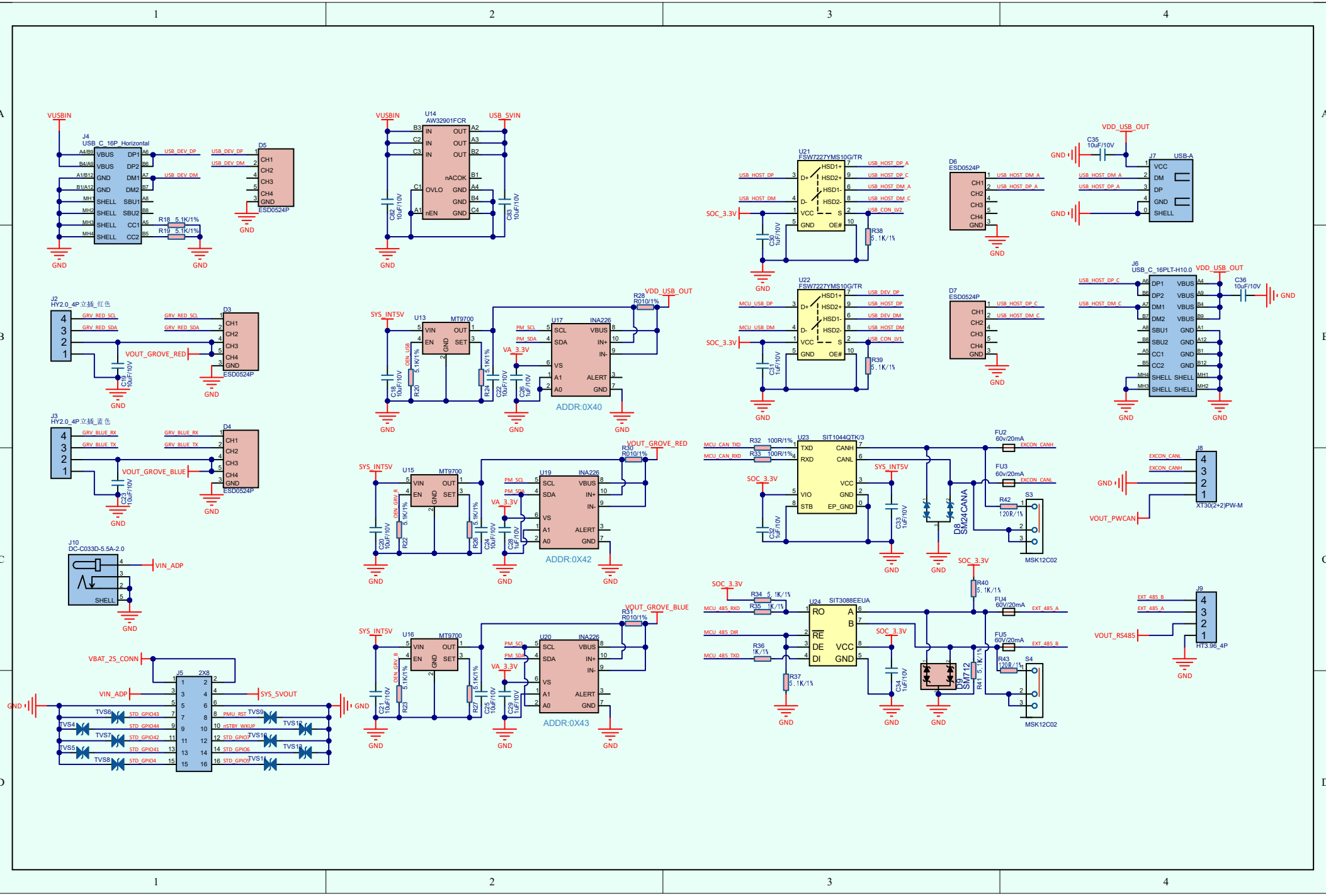
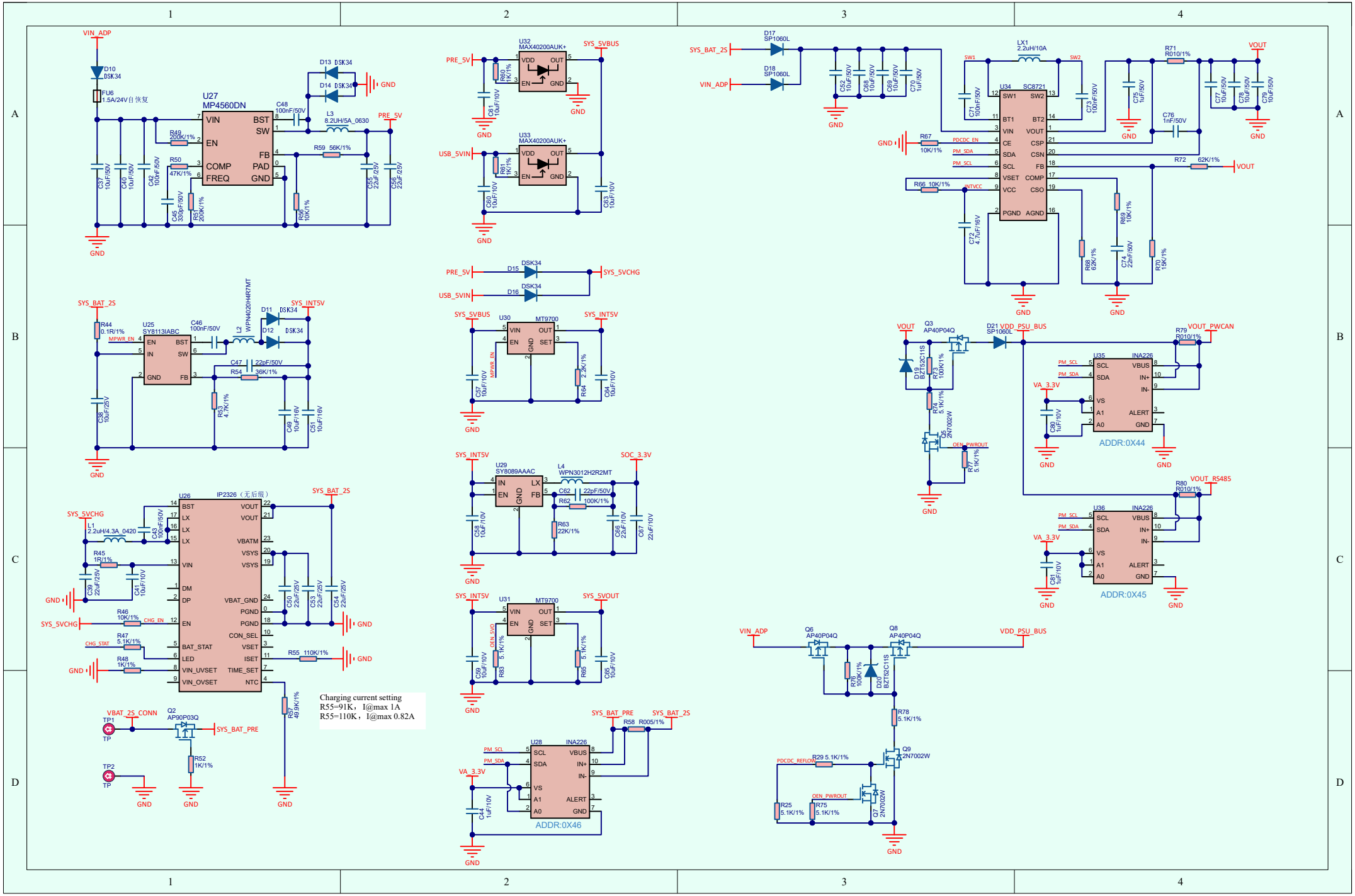

U5
ESP32-S3-WROOM-1U-N16R8
Reserved for N16R8
U11
STM32G031G8U6
U12
RX8130CE
J1
CON5
PMU_KEY2
USER KEY
U1
U3
U6
U7
U8
U9
WS2812C_2020
TP7
TP8
TP9
TP10
BATTERY_RTC


1
2
3
4
A
B
C
D
VUSBIN
J4
USB_C_16P_Horizontal
VBUS
GND
SHELL
USB_DEV_DP
USB_DEV_DM
R18_5.1K/1%
R19_5.1K/1%
D5
ESD0524P
U14
AW32901FCR
IN
OUT
nACKOK
OVLO
nEN
USB_5VIN
C82 10uF/10V
C83 10uF/10V
VDD_USB_OUT
U21
FSW7227YMS10G/TR
HSD1+
HSD2+
HSD1-
HSD2-
D+
D-
VCC
OE#
USB_HOST_DP
USB_HOST_DM
SOC_3.3V
C30 1uF/10V
R38 5.1K/1%
USB_HOST_DP_A
USB_HOST_DM_A
USB_HOST_DP_C
USB_HOST_DM_C
USB_CON_LV2
D6
ESD0524P
C35 10uF/10V
J7
USB-A
VCC
DM
DP
GND
SHELL
J2
HY2.0_4P立插_红色
GRV_RED_SCL
GRV_RED_SDA
VOUT_GROVE_RED
C19 10uF/10V
D3
ESD0524P
SYS_INT5V
U13
MT9700
VIN
EN
SET
OEN_USB
R20 5.1K/1%
C18 10uF/10V
R24 5.1K/1%
C22 10uF/10V
U17
INA226
SCL
SDA
VS
A1
A0
VBUS
IN+
IN-
ALERT
PM_SCL
PM_SDA
VA_3.3V
C28 1uF/10V
ADDR:0X40
R28 R010/1%
VOUT_GROVE_RED
U22
FSW7227YMS10G/TR
MCU_USB_DP
MCU_USB_DM
C31 1uF/10V
R39 5.1K/1%
USB_CON_LV1
D7
ESD0524P
J6
USB_C_16PLT-H10.0
DP1
DP2
DM1
DM2
SBU1
SBU2
CC1
CC2
SHELL
C36 10uF/10V
J3
HY2.0_4P立插_蓝色
GRV_BLUE_RX
GRV_BLUE_TX
VOUT_GROVE_BLUE
C21 10uF/10V
D4
ESD0524P
U15
MT9700
OEN_GRV_R
R22 5.1K/1%
C20 10uF/10V
R26 5.1K/1%
C24 10uF/10V
U19
INA226
C29 1uF/10V
ADDR:0X42
R30 R010/1%
VOUT_GROVE_RED
U23
SIT1044QTK/3
MCU_CAN_TXD
MCU_CAN_RXD
R32 100R/1%
R33 100R/1%
TXD
RXD
CANH
CANL
VIO
STB
EP_GND
C32 1uF/10V
C33 1uF/10V
D8
SM24CANA
FU2
60v/20mA
FU3
60v/20mA
EXCON_CANH
EXCON_CANL
R42 120R/1%
S3
MSK12C02
J8
XT30(2+2)PW-M
VOUT_PWCAN
J10
DC-C033D-5.5A-2.0
VIN_ADP
U16
MT9700
OEN_GRV_B
R23 5.1K/1%
C23 10uF/10V
R27 5.1K/1%
C25 10uF/10V
U20
INA226
C29 1uF/10V
ADDR:0X43
R31 R010/1%
VOUT_GROVE_BLUE
U24
SIT3088EEUA
RO
RE
DE
DI
A
B
VCC
GND
MCU_485_RXD
MCU_485_DIR
MCU_485_TXD
R34 5.1K/1%
R35 1K/1%
R36 1K/1%
R37 5.1K/1%
C34 1uF/10V
R40 5.1K/1%
FU4
60v/20mA
FU5
60v/20mA
EXT_485_A
EXT_485_B
R43 120R/1%
R41 5.1K/1%
D9
SM712
S4
MSK12C02
J9
HT3.96_4P
VOUT_RS485
VBAT_2S_CONN
VIN_ADP
SYS_5VOUT
J5
2X8
TVS4
TVS5
TVS6
TVS7
TVS8
TVS9
TVS10
TVS11
TVS12
TVS13
TVS14
TVS15
TVS16
STD_GPIO43
STD_GPIO44
STD_GPIO42
STD_GPIO41
STD_GPIO4
PMU_RST
nSTBY_WKUP
STD_GPIO5
STD_GPIO6
STD_GPIO7


VIN_ADP
D10 DSK34
FU6 1.5A/24V自恢复
U27 MP4560DN
VIN
EN
COMP
FREQ
BST
SW
FB
PAD
GND
D13 DSK34
D14 DSK34
L3 8.2UH/5A_0630
PRE_5V
R49 200K/1%
R50 47K/1%
R59 56K/1%
C48 100nF/50V
C37 10uF/50V
C40 10uF/50V
C42 100nF/50V
C45 330pF/50V
R51 200K/1%
R56 10K/1%
C55 22uF/25V
C56 22uF/25V
U32 MAX40200AUK+
U33 MAX40200AUK+
SYS_5VBUS
USB_5VIN
R60 1K/1%
R61 1K/1%
C61 10uF/10V
C60 10uF/10V
C63 10uF/10V
D15 DSK34
D16 DSK34
SYS_5VCHG
SYS_BAT_2S
U25 SY8113IABC
MPWR_EN
L2 WPN4020H4R7MT
D11 DSK34
D12 DSK34
SYS_INT5V
R44 0.1R/1%
C46 100nF/50V
C47 22pF/50V
R54 36K/1%
R53 4.7K/1%
C38 10uF/25V
C49 10uF/16V
C51 10uF/16V
U30 MT9700
C57 10uF/10V
R64 2.2K/1%
C64 10uF/10V
U26 IP2326（无后缀）
L1 2.2uH/4.3A_0420
C43 100nF/50V
R45 1R/1%
C39 22uF/25V
C41 10uF/10V
R46 10K/1%
CHG_EN
R47 5.1K/1%
CHG_STAT
R48 1K/1%
C50 22uF/25V
C53 22uF/25V
C54 22uF/25V
R55 110K/1%
R57 49.9K/1%
Charging current setting
R55=91K，I@max 1A
R55=110K，I@max 0.82A
VBAT_2S_CONN
TP1
TP2
Q2 AP90P03Q
SYS_BAT_PRE
R52 1K/1%
U29 SY8089AAAC
L4 WPN3012H2R2MT
SOC_3.3V
C62 22pF/50V
R62 100K/1%
R63 22K/1%
C58 10uF/10V
C66 22uF/10V
C67 22uF/10V
U31 MT9700
SYS_5VOUT
C59 10uF/10V
R83 5.1K/1%
R65 5.1K/1%
C65 10uF/10V
U28 INA226
R58 R005/1%
PM_SCL
PM_SDA
VA_3.3V
C44 1uF/10V
ADDR:0X46
D17 SP1060L
D18 SP1060L
SYS_BAT_2S
VIN_ADP
C52 10uF/50V
C68 10uF/50V
C69 10uF/50V
C70 1uF/50V
LX1 2.2uH/10A
U34 SC8721
SW1
SW2
BT1
BT2
VIN
VOUT
CE
CSP
SDA
CSN
SCL
FB
VSET
COMP
VCC
CSO
PGND
AGND
C71 100nF/50V
C73 100nF/50V
C75 1uF/50V
C76 1nF/50V
C72 4.7uF/16V
R71 R010/1%
R72 62K/1%
R67 10K/1%
R66 10K/1%
PDCDC_EN
INTVCC
R68 62K/1%
R69 10K/1%
R70 15K/1%
C74 22nF/50V
C77 10uF/50V
C78 10uF/50V
C79 10uF/50V
VOUT
Q3 AP40P04Q
D21 VDD_PSU_BUS
SP1060L
D19 BZT52C11S
R73 100K/1%
R74 5.1K/1%
Q5 2N7002W
OEN_PWROUT
R77 5.1K/1%
R79 R010/1%
VOUT_PWCAN
U35 INA226
C80 1uF/10V
ADDR:0X44
R80 R010/1%
VOUT_RS485
U36 INA226
C81 1uF/10V
ADDR:0X45
VDD_PSU_BUS
Q6 AP40P04Q
Q8 AP40P04Q
R76 100K/1%
D20 BZT52C11S
R78 5.1K/1%
Q9 2N7002W
PDCDC_REFLOW
R29 5.1K/1%
R25 5.1K/1%
R75 5.1K/1%
Q7 2N7002W
GND